

Question	Marking Guidance	Mark	Comments
10.1	1. Reduction in ATP production by aerobic respiration; 2. Less force generated because fewer actin and myosin interactions in muscle; 3. Fatigue caused by lactate from anaerobic respiration;	3	
10.2	Couple A, 1. Mutation in mitochondrial DNA/DNA of mitochondrion affected; 2. All children got affected mitochondria from mother; 3. (Probably mutation) during formation of mother's ovary/eggs; Couple B, 4. Mutation in nuclear gene/DNA in nucleus affected; 5. Parents heterozygous; 6. Expect 1 in 4 homozygous affected;	4 max	
10.3	1. Change to tRNA leads to wrong amino acid being incorporated into protein; 2. Tertiary structure (of protein) changed; 3. Protein required for oxidative phosphorylation/the Krebs cycle, so less/no ATP made;	3	
10.4	1. Mitochondria/aerobic respiration not producing much/any ATP; 2. (With MD) increased use of ATP supplied by increase in anaerobic respiration; 3. More lactate produced and leaves muscle by (facilitated) diffusion;	3	
10.5	1. Enough DNA using PCR; 2. Compare DNA sequence with 'normal' DNA;	2	

